```
class TicketBooking {
    String passengerName;
    String bookingStatus;

    TicketBooking(String passengerName) {
        this.passengerName = passengerName;
        this.bookingStatus = "Booked";
    }

    void processBooking() {
        bookingStatus = "Confirmed";
    }

    void displayBookingDetails() {
        System.out.println("Passenger Name: " +
passengerName);
        System.out.println("Booking Status: " +
bookingStatus);
        System.out.println("Thread Name: " +
Thread.currentThread().getName());
    }
}

class BookingThread extends Thread {
    TicketBooking booking;
    boolean displayDetails;

    BookingThread(TicketBooking booking) {
        this(booking, true);
    }

    BookingThread(TicketBooking booking, boolean
displayDetails) {
        this.booking = booking;
        this.displayDetails = displayDetails;
    }

    public void run() {
        if (booking == null) {
            return;
        }

        booking.processBooking();

        if (displayDetails) {
            booking.displayBookingDetails();
        }
    }
}

class BookingRunnable implements Runnable {
    TicketBooking booking;
    boolean displayDetails;

    BookingRunnable(TicketBooking booking) {
        this(booking, true);
    }

    BookingRunnable(TicketBooking booking, boolean
displayDetails) {
        this.booking = booking;
        this.displayDetails = displayDetails;
    }

    public void run() {
        if (booking == null) {
            return;
        }

        booking.processBooking();

        if (displayDetails) {
            booking.displayBookingDetails();
        }
    }
}
```

```
public class pro8 {
    public static void main(String[] args) {

        TicketBooking booking1 = new
TicketBooking("Rahul");
        TicketBooking booking2 = new
TicketBooking("Priya");
        TicketBooking booking3 = new
TicketBooking("Aman");
        TicketBooking booking4 = new
TicketBooking("Neha");
        TicketBooking booking5 = new
TicketBooking("Arjun");

        BookingThread thread1 = new
BookingThread(booking1);
        BookingThread thread2 = new
BookingThread(booking2);

        BookingRunnable runnable1 = new
BookingRunnable(booking3);
        BookingRunnable runnable2 = new
BookingRunnable(booking4);

        Thread thread3 = new Thread(runnable1);
        Thread thread4 = new Thread(runnable2);

        Thread thread5 = new Thread(new
BookingRunnable(booking5));

        thread1.start();
        thread2.start();
        thread3.start();
        thread4.start();
        thread5.start();

        try {
            thread1.join();
            thread2.join();
            thread3.join();
            thread4.join();
            thread5.join();
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        TicketBooking emptyBooking = new
TicketBooking("");
        TicketBooking nullNameBooking = new
TicketBooking(null);

        Thread emptyNameThread = new
BookingThread(emptyBooking);
        Thread nullNameThread = new
BookingThread(nullNameBooking);

        emptyNameThread.start();
        nullNameThread.start();

        try {
            emptyNameThread.join();
            nullNameThread.join();
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        BookingThread invalidBookingThread = new
BookingThread(null);
        invalidBookingThread.start();

        try {
            invalidBookingThread.join();
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        TicketBooking sharedBooking = new
TicketBooking("Shared Passenger");

        Thread sharedThread1 = new Thread(new
BookingRunnable(sharedBooking));
        Thread sharedThread2 = new Thread(new
BookingRunnable(sharedBooking));
```

```
        sharedThread1.start();
        sharedThread2.start();

        try {
            sharedThread1.join();
            sharedThread2.join();
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        Thread interruptedThread = new Thread(new
Runnable() {
            public void run() {
                try {
                    Thread.sleep(1000);
                    TicketBooking booking = new
TicketBooking("Interrupted Passenger");
                    booking.processBooking();
                    booking.displayBookingDetails();
                } catch (InterruptedException e) {
                    Thread.currentThread().interrupt();
                }
            }
        });

        interruptedThread.start();
        interruptedThread.interrupt();

        try {
            interruptedThread.join();
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        Thread[] largeBookings = new Thread[20];

        for (int i = 0; i < 20; i++) {
            largeBookings[i] =
                new BookingThread(
                    new TicketBooking("Passenger" + (i + 1)),
                    true
                );

            largeBookings[i].start();
        }

        try {
            for (int i = 0; i < 20; i++) {
                largeBookings[i].join();
            }
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }

        Thread[] excessiveBookings = new Thread[500];

        for (int i = 0; i < 500; i++) {
            excessiveBookings[i] =
                new BookingThread(
                    new TicketBooking("BulkPassenger" + (i +
1)),
                    false
                );

            excessiveBookings[i].start();
        }

        try {
            for (int i = 0; i < 500; i++) {
                excessiveBookings[i].join();
            }
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }
```

```
        System.out.println(
            "All passenger booking requests have been processed successfully."
        );
    }
}
```

Passenger Name: Priya

Passenger Name: Aman

Booking Status: Confirmed

Passenger Name: Rahul

Passenger Name: Arjun

Booking Status: Confirmed

Thread Name: Thread-4

Passenger Name: Neha

Booking Status: Confirmed

Thread Name: Thread-3

Booking Status: Confirmed

Thread Name: Thread-2

Booking Status: Confirmed

Thread Name: Thread-0

Thread Name: Thread-1

Passenger Name:

Booking Status: Confirmed

Thread Name: Thread-5

Passenger Name: null

Booking Status: Confirmed

Thread Name: Thread-6

Passenger Name: Shared Passenger

Booking Status: Confirmed

Thread Name: Thread-8

Passenger Name: Shared Passenger

Booking Status: Confirmed

Thread Name: Thread-9

Passenger Name: Passenger1

Passenger Name: Passenger2

Passenger Name: Passenger6

Booking Status: Confirmed

Thread Name: Thread-16

Passenger Name: Passenger3

Booking Status: Confirmed

Booking Status: Confirmed

Thread Name: Thread-11

Passenger Name: Passenger14

Booking Status: Confirmed

Thread Name: Thread-24

Passenger Name: Passenger4

Booking Status: Confirmed

Thread Name: Thread-14

Booking Status: Confirmed

Passenger Name: Passenger18

Booking Status: Confirmed

Thread Name: Thread-28

Passenger Name: Passenger5

Booking Status: Confirmed

Thread Name: Thread-15

Passenger Name: Passenger7

Booking Status: Confirmed

Thread Name: Thread-17

Passenger Name: Passenger8

Booking Status: Confirmed

Thread Name: Thread-18

Thread Name: Thread-13

Passenger Name: Passenger9

Booking Status: Confirmed

Thread Name: Thread-19

Passenger Name: Passenger11

Booking Status: Confirmed

Thread Name: Thread-21

Passenger Name: Passenger10

Booking Status: Confirmed

Thread Name: Thread-20

Passenger Name: Passenger13

Booking Status: Confirmed

Thread Name: Thread-23

Passenger Name: Passenger12

Booking Status: Confirmed

Thread Name: Thread-22

Passenger Name: Passenger16

Passenger Name: Passenger17

Passenger Name: Passenger19

Booking Status: Confirmed

Thread Name: Thread-29

Passenger Name: Passenger20

Booking Status: Confirmed

Passenger Name: Passenger15

Thread Name: Thread-12

Booking Status: Confirmed

Thread Name: Thread-26

Booking Status: Confirmed

Thread Name: Thread-30

Booking Status: Confirmed

Thread Name: Thread-25

Thread Name: Thread-27